

Particle properties of waves
Part -6
Problems on Compton Effect

1. Calculate the minimum energy required by a photon to transfer half of its energy to an electron at rest.

Solution :

Here we have to find the minimum energy that means the frequency is also minimum, so the wavelength will be maximum. Maximum wavelength will occur when $\theta = 180^\circ$

$$E = h\nu$$
$$\lambda' - \lambda = \lambda_c(1 - \cos\theta)$$
$$\lambda' - \lambda = 2\lambda_c \longrightarrow (1)$$

According to the question $E_{initial} = h\nu$

$$E_{final} = h\nu' = \frac{h\nu}{2}$$

$$\frac{hc}{\lambda'} = \frac{hc}{2\lambda}$$

$$\lambda' = 2\lambda, \text{ Substituting this value in (1)}$$

$$2\lambda - \lambda = 2\lambda_c$$

$$\lambda = 2\lambda_c = 2 \times 2.426 \text{ pm}$$

$$E_{min} = \frac{hc}{\lambda} = \frac{1.26 \times 10^{-6}}{2 \times 2.426} = 25 \times 10^4 \text{ eV} = 0.25 \text{ MeV}$$

2. The kinetic energy of the recoil electron if a photon of wavelength $2.5\text{\AA}$ scattered at 90° is -----

- (a) 40 eV
- (b) 47.8 eV
- (c) 88 eV
- (d) 24.6 eV

Solution : (b) 47.8 eV

Given

$$\lambda = 2.5\text{\AA} = 250 \times 10^{-17} \text{ m}$$

$$\theta = 90^\circ$$

$$K = ?$$

$$\lambda' - \lambda = \lambda_c$$

$$\lambda' = \lambda + \lambda_c$$

$$\lambda' = 250 \text{ pm} + 2.626 \text{ pm} = 252.4 \text{ pm}$$

$$K = h\nu - h\nu'$$

$$K = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right) = \left(\frac{\lambda' - \lambda}{\lambda \lambda'} \right)$$

$$= 1.26 \times 10^{-6} \text{ eV} \left(\frac{2.426 \text{ pm}}{250 \text{ pm} \times 252.4 \text{ pm}} \right)$$

$$K = 48 \text{ eV} \approx 47.8 \text{ eV}$$

3. Calculate the maximum percentage change in wavelength of $10\text{\AA}$ incident photons in Compton scattering.

Solution :

Given $\lambda = 10\text{\AA} = 10 \times 10^{-10} \text{ m}$

$$\frac{\Delta\lambda}{\lambda} \times 100\% = ?$$

$$\Delta\lambda = 2\lambda_c$$

For λ to be maximum $\theta = 180^\circ$

$$\frac{\Delta\lambda}{\lambda} \times 100\% = \frac{2 \times 2.426 \times 10^{-12}}{10 \times 10^{-10}} \times 100\% = 0.484\%$$

4. A beam of X-rays of wavelength 0.2 nm is incident on a free electron and gets scattered in a direction with respect to the direction of the incident radiation resulting in maximum wavelength shift. Find the percentage energy loss of the incident radiation

(a) 4.8%

(b) 6.2%

(c) 2.4%

(d) 1.2%

Solution : (c) 2.4%

Given $\lambda = 0.2 \text{ nm} = 200 \text{ pm}$

$$\lambda' - \lambda = \text{maximum} = 2\lambda_c \quad (\theta = 180^\circ)$$

$$\frac{\Delta E}{E} \times 100 \% = ?$$

$$\begin{aligned}\lambda' &= \lambda + 2\lambda_c \\ &= (200 + 4.852) \text{ pm} = 204.8 \text{ pm}\end{aligned}$$

$$\begin{aligned}\frac{\Delta E}{E} \times 100 \% &= \frac{h\nu - h\nu'}{h\nu} = \frac{\frac{c}{\lambda} - \frac{c}{\lambda'}}{\frac{c}{\lambda}} \\ &= \left(\frac{\lambda' - \lambda}{\lambda \lambda'} \right) \times \lambda \times 100\% \\ &= \left(\frac{\lambda' - \lambda}{\lambda'} \right) \times 100\%\end{aligned}$$

$$\frac{\Delta E}{E} \times 100 \% = \frac{2 \times 2.426 \text{ pm}}{204.8 \text{ pm}} \times 100\% = 2.4\%$$

5. X-rays of $4 \text{ \AA}$ wavelength falls on an electron cloud and gets scattered.

Determine the maximum change in

- i) Kinetic energy of e-
- ii) Velocity of e-

Solution :

Given $\lambda = 4 \text{ \AA} = 400 \text{ pm}$

i) Maximum change in kinetic energy occur only when $\theta = 180^\circ$

$$\lambda - \lambda' = 2\lambda_c = 4.852 \text{ pm} \approx 5 \text{ pm}$$

$$\lambda' = 400 + 4.852 = 404.852 \text{ pm} \approx 405 \text{ pm}$$

$$\begin{aligned}K &= hc \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right) \\ &= hc \left(\frac{\lambda' - \lambda}{\lambda \lambda'} \right) = \frac{1.24 \times 10^{-6} \times 5}{400 \text{ pm} \times 405 \text{ pm}}\end{aligned}$$

$$K = 37 \text{ eV}$$

ii) $K = \frac{1}{2} mv^2$

$$V = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2 \times 37 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}} = 3.6 \times 10^6 \text{ m/s}$$